Thinking With Containers: A Multi-Agent Retrieval Approach for the Case-Based Semantic Search of Architectural Designs

Viktor Ayzenshtadt[1,2], Christoph Langenhan[3], Saqib Bukhari[2],
Klaus-Dieter Althoff[1,2], Frank Petzold[3] and Andreas Dengel[2]

[1]*University of Hildesheim, Institute of Computer Science, Samelsonplatz 1, 31141 Hildesheim, Germany*

[2]*German Research Center for Artificial Intelligence (DFKI), Trippstadter Strae 122, 67663 Kaiserslautern, Germany*

[3]*Faculty of Architecture, Technical University of Munich, Arcisstrasse 21, 80333 Munich, Germany*

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Abstract: To provide the retrieval of information, that can be considered useful during the design conceptualization process, with advantages of distributed artificial knowledge, an approach, that distributes retrieval-related and knowledge maintaining tasks among autonomously working and case-based self-learning agents and agent groups, can be used. In this work we present the distributed retrieval system MetisCBR for the architectural design domain, where agents work in groups (containers) on resolving of user queries built with a semantic description model Semantic Fingerprint. The main aim of our approach is to carry out a basis for a considerable retrieval tool for architects, where the combination of case-based reasoning and multi-agent methods helps to achieve valuable and helpful search results in a comprehensive building design collection.

1 INTRODUCTION

The early conceptualization phase of a building is the stage of the architectural design development, during which an architect can increase the efficiency of the working process by communicating with a computer-aided helper system. Such a system should be able to present new ideas and helpful recommendations for inspiration and guidance in combination with similar building designs to a currently composed one.

The study of buildings that have a similar context to the design problem at hand, or that are based on a similar initial premise, is seen as a way of approaching a design problem and developing a possible course of action. Prior built projects or architectural designs serve here as a knowledge base that contains both examples of spatial constellations as well as solutions for specific architectural situations.

Plans, models, and the documentation of architectural designs and existing buildings and urban environments represent an extensive architectural knowledge base which contains implicit design knowledge that is valuable for the design process. The problem is, that this knowledge is not available in an explicit form, without first analyzing and interpreting the information available.

To handle this architectural knowledge, for the purpose of providing a retrieval component that efficiently interprets this data, the helper systems often use *case-based reasoning* (CBR) methods as the underlying retrieval technique. This aims to achieve high concentration on the quality of results, and to make the system knowledge-intensive and extensible.

For implementation of such retrieval features, many approaches were proposed and applied to the existing systems in the last three decades. The most significant of them are listed in (Heylighen and Neuckermans, 2001) and (Richter et al., 2007).

In this work we present *MetisCBR*, a multi-agent and case-based retrieval approach for similar architectural designs. This approach uses inter alia case-based learning agents as retrieval units in a *distributed container-based* application. MetisCBR was implemented as a prototype and integrated into the existing infrastructure in context of a basic research project *Metis – Knowledge-based search and query methods for the development of semantic information models (BIM) for use in early design phases*, an interdisciplinary project of CBR, *multi-agent systems* (MAS), and *computer-aided architectural design* (CAAD), partially funded by the German Research Foundation (Deutsche Forschungsgemeinschaft, DFG).

This paper is structured as follows: first, the background and the interdisciplinarity of the research

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基于容器的思维：一种用于建筑设计案例语义搜索的多智能体检索方法

维克托·艾曾施塔特[1,2]，克里斯托夫·朗根汉[3]，萨基布·布哈里[2]，克劳斯-迪特尔·阿尔托夫[1,2]，弗兰克·佩茨奥尔德[3]以及安德烈亚斯·登格尔[2]

[1]*University of Hildesheim, Institute of Computer Science, Samelsonplatz 1, 31141 Hildesheim, Germany*

[2]*German Research Center for Artificial Intelligence (DFKI), Trippstadter Strae 122, 67663 Kaiserslautern, Germany*

[3]*Faculty of Architecture, Technical University of Munich, Arcisstrasse 21, 80333 Munich, Germany*

关键词：多智能体系统、基于案例的智能体、基于案例的检索、设计支持。

摘要：　为了在设计概念化过程中提供可被视为有用的信息检索，并利用分布式人工智能知识的优势，可以采用一种将检索相关任务与知识维护任务分散至自主运行且基于案例的自学习智能体及智能体群体的方法。在本研究中，我们提出了面向建筑设计领域的分布式检索系统MetisCBR，该系统中的智能体以群体（容器）为单位，基于语义描述模型“语义指纹”来处理用户查询。我们方法的主要目标是为建筑师构建一个功能强大的检索工具，通过将基于案例的推理与多智能体方法相结合，在庞大的建筑设计资源库中实现有价值且实用的搜索结果。

1 引言

建筑的早期概念设计阶段是建筑设计开发过程中的一个重要环节，在此阶段，建筑师可通过与计算机辅助辅助系统进行交互来提升工作流程的效率。该系统应能够结合当前正在设计的建筑方案，呈现新的设计理念，并提供具有启发性和指导性的参考建议，以相似的既有建筑案例作为支撑。

研究与当前设计问题具有相似背景或基于相同初始假设的建筑，被视为一种接近设计问题并制定可行行动方案的途径。既有的建成项目或建筑设计在此充当知识库，其中既包含空间布局的范例，也提供针对特定建筑情境的解决方案。

建筑方案、模型以及建筑设计和既有建筑与城市环境的文档，共同构成了一个庞大的建筑知识库，其中蕴含着对设计过程具有重要价值的隐性设计知识。然而，问题在于，这些知识并未以显式的形式直接呈现，必须先对现有信息进行分析与解读后方可获取。

为了处理这一架构知识，对于为构建一个能够高效解析此类数据的检索组件，辅助系统通常采用基于案例的推理（CBR）方法作为其底层检索技术。此举旨在确保结果质量的高度聚焦，并使系统具备知识密集性和可扩展性。

为实现此类检索功能，过去三十年中已有多种方法被提出并应用于现有系统。其中最为重要的方法已列于Heylighen和Neuckermans（2001）以及Richter等人（2007）的文献中。

在本研究中，我们提出了*MetisCBR*，一种用于相似建筑设计的多智能体与基于案例的检索方法。该方法以基于案例的学习智能体作为检索单元，并将其应用于*distributed container-based*场景。MetisCBR被实现为一个原型，并集成到现有基础设施中，作为一项基础研究项目*Metis – Knowledge-based search and query methods for the development of semantic information models (BIM) for use in early design phases*的一部分；该项目是一个跨学科研究项目，涉及基于案例推理（CBR）、*multi-agent systems* (多智能体系统（MAS）以及*computer-aided architectural design* (计算机辅助建筑设计（CAAD），并由德国研究联合会（Deutsche Forschungsgemeinschaft, DFG）部分资助。

本文的结构安排如下：首先，介绍研究的背景及其跨学科特性。

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project *Metis* will be briefly presented. The next section contains the overview of related approaches and studies. Next, a short overview of the system structure will be presented. The next section characterizes the proposed agent types and their corresponding tasks in detail. In the next section the prototype of the approach and its context and validation within the research project will be presented. The conclusion summarizes the work and the future project planning.

2 BACKGROUND

Technological developments and globalized working processes have transformed the building design process. However, current digital semantic building models are no longer able to adequately represent the increasing complexity of modern architectural projects. The potential of combining agent-based, case-based, and graph-based methods with novel semantic models, for example, for energy calculations or spatial research strategies, is not fully exploited. To find essential ways to utilize this potential, a basic research project *Metis* was initiated. The distributed CBR retrieval system *MetisCBR* we present in this work is one of the integrant parts of the project. The interdisciplinary nature of the project provides a necessary mixture to overcome the problem of timely finding similar building designs as source of inspiration to guide the design process of buildings.

CAAD considers computer-related topics regarding buildings and delivers insights into ways an architect can work/interact with such a system or how she would evaluate search results.

CBR, as the discipline responsible for working with intensive knowledge data, is the means for the actual finding of similar building designs. CBR's all-purpose nature and a multitude of different CBR retrieval approaches provide efficient tools for the accomplishment of the project's main tasks.

MAS play a role of executing units in the project. Being a means for distributed decentralized problem solving, MAS, and the corresponding research area, provide a necessary collection of techniques for distributed and concurrent retrieval processes.

To combine the above named fields in order to work together in a common system, a *building information model (BIM) infrastructure* was built for the project. It integrates every project-related service, including a building model server, a graph database with building designs represented as graphs, a content management system, and software prototypes.

3 RELATED WORK

The system we present in this work is set in context of the above mentioned research project *Metis*, and is related to the CAAD and BIM applications with case-based reasoning as the underlying retrieval technique.

The project-related work includes (Siebert, 2014), a research work that contains the initial setting of our system where a concept of agent-based experience sharing system is applied to the architectural domain.

A semantic building structure description model *Semantic Fingerpint* (Langenhan and Petzold, 2010) was initiated to provide a structure to idea that building floor plans can be described by abstracting the building data at the level of building, storey, and room, as well as their respective topological relationships. Based on this model, a subgraph matching-based retrieval algorithm was proposed in (Ahmed et al., 2014), where an information extraction process of floor plans is also contained and later used in the *Metis* project for the graph database.

Applications for Android and iPad, and a touch-table interface, were developed for query construction with modern haptic HCI methods. A web-based user interface (Bayer et al., 2015) is part of the project as well. This application uses the *(AGraphML)* specification (Langenhan, 2015) for the query construction.

Finally, a distributed domain model that builds a CBR base for the retrieval process of *MetisCBR* was introduced in (Ayzenshtadt et al., 2015).

CBR-based approaches with architectural emphasis can be classified as a subdomain of *case-based design* (CBD). In these systems, the retrieval functionality is one of the most essential system modules.

For example, in FABEL (Voss, 1997) functional entities called *specialists* are used to execute the retrieval tasks. The *specialists* work with special *aspect-specific* representations of cases and queries. These representations are produced by applying a transformation process to the user query or to a currently comparing case of the case base. *Retrieval specialists* determine similar cases by comparing the corresponding aspects of cases and queries.

CBArch (Cavieres et al., 2011) is a supporting framework for the conceptualization phase of architectural design. With emphasis on the early energy performance evaluation of buildings with commercial background, it provides an approach for a completely CBR-based solution for such cases.

Other systems, such as DYNAMO (Heylighen et al., 2002), SEED (Flemming, 1994), PRECEDENTS (Oxman and Oxman, 1993) or CaseBook (Inanc, 2000), can also be mentioned as closely related, as they serve the same purpose as our system.

项目*Metis*将作简要介绍。下一节概述相关方法与研究。随后，将对系统架构进行简要说明。接下来的章节将详细阐述所提出的智能体类型及其相应任务。在下一节中，将介绍该方法的原型，以及其在本研究项目中的应用背景与验证情况。最后，结论部分对本文工作及未来项目规划进行了总结。

2 背景

技术进步与全球化的工作流程已深刻改变了建筑设计过程。然而，现有的数字化语义建筑模型已难以充分表征现代建筑项目的日益复杂性。例如，在能源计算或空间研究策略等方面，将基于智能体、基于案例以及基于图的方法与新型语义模型相结合的潜力尚未得到充分挖掘。为探寻有效利用这一潜力的关键途径，一项基础性研究项目*Metis*已被启动。本文所提出的分布式CBR检索系统*MetisCBR*是该项目的重要组成部分之一。该项目的跨学科特性提供了必要的协同机制，以克服在设计过程中及时寻找相似建筑方案这一难题，从而为建筑设计过程提供灵感与指导。

CAAD 该研究探讨了与建筑相关的计算机技术，并深入分析了建筑师如何与这类系统协作或交互，以及她将如何评估搜索结果。

*CBR*作为一门以处理密集型知识数据为己任的学科，案例推理是发现相似建筑设计的有效途径。案例推理的通用性及其多种检索方法为实现项目的主要目标提供了高效的工具。

MAS 在项目中发挥执行单元的作用。作为分布式、去中心化问题求解的手段，多智能体系统及其相关研究领域为分布式并行检索过程提供了必要的技术集合。

为将上述各领域整合至一个统一的系统中协同工作，本项目构建了一个*building information model (BIM) infrastructure*。该系统集成了所有与项目相关的服务，包括建筑模型服务器、以图结构表示建筑方案的图数据库，以及内容管理系统和软件原型。

3 相关工作

本文所提出的系统置于上述研究项目*Metis*的背景下，与基于案例推理作为底层检索技术的计算机辅助建筑设计（CAAD）及建筑信息建模（BIM）应用相关。

与该项目相关的工作包括（Siebert，2014），这是一项研究工作，其中首次构建了我们的系统，并将基于智能体的经验共享机制应用于建筑领域。

Langenhan和Petzold于2010年提出了一种语义建筑结构描述模型*Semantic Fingerpint* (，旨在为以下观点提供框架：可通过在建筑、楼层及房间层面抽象建筑数据，并刻画其相应的拓扑关系，来描述建筑平面图。基于该模型，Ahmed等人于2014年进一步提出了一种基于子图匹配的检索算法，其中还包含一个用于提取平面图信息的处理流程，并随后在*Metis*项目中应用于图数据库。

针对Android和iPad的应用程序以及触控桌界面，已采用现代触觉人机交互方法开发用于查询构建。此外，该项目还包含一个基于Web的用户界面（Bayer等，2015）。该应用程序在查询构建过程中使用*(AGraphML)*规范（Langenhan，2015）。

最后，Ayzenshtadt 等人（2015）提出了一种分布式领域模型，该模型为 *MetisCBR* 的检索过程构建了一个基于案例的推理库。

以体系结构为重点的基于案例推理的方法可被归类为*case-based de-sign* (CBD)的一个子领域。在这些系统中，检索功能是最为重要的系统模块之一。

例如，在FABEL系统中（Voss，1997），被称为*specialists*的功能实体用于执行检索任务。这些*specialists*与案例和查询的特定*aspect-specific*表示形式协同工作。这些表示形式是通过对用户查询或案例库中正在比较的当前案例施加转换过程而生成的。*Retrieval spe- cialists*通过比较案例与查询的相应方面来确定相似案例。

CBArch（Cavieres 等，2011）是一个用于建筑设计概念化阶段的支撑框架。该框架重点关注具有商业背景建筑的早期能源性能评估，并为这类场景提供了一种完全基于案例推理（CBR）的解决方案。

其他系统，如DYNAMO（Heylighen等，2002）、SEED（Flemming，1994）、PRECE- DENTS（Oxman和Oxman，1993）以及CaseBook（Inanc，2000），也可被视为与其密切相关，因为它们与本系统具有相同的用途。

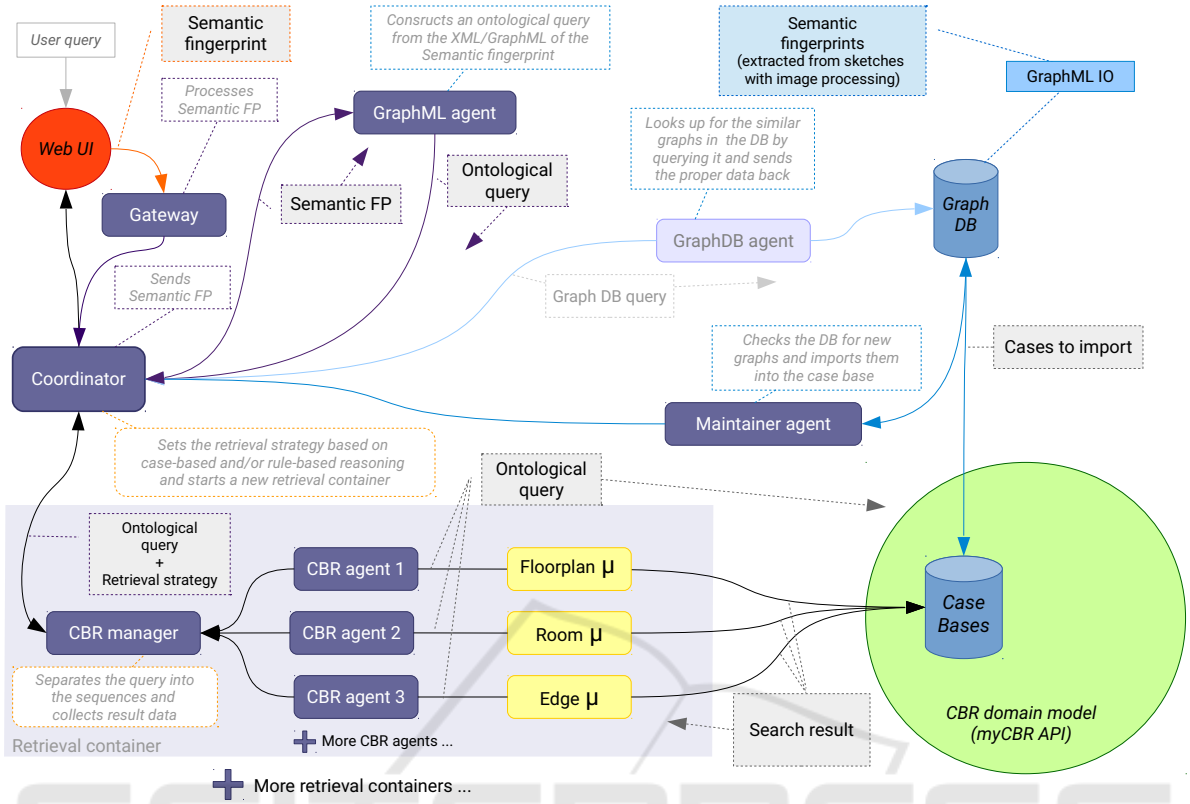


Figure 1: The system overview of MetisCBR.

4 THE SYSTEM OVERVIEW

In this section we provide a brief overview of the proposed approach *MetisCBR*. The main task of the entire multi-agent system is to process a query in the AGraphML format that was sent by a user *(an architect)* from a web-based query construction interface (not part of this approach, see also Sections 3 and 6).

As depicted in Figure 1, the system is constructed in a circular form, where the *gateway* marks the entry point into the system, after a user query was sent from the web interface.

Successfully passing the gateway agent that conducts an initial error analysis, the query reaches the *coordination component* that is responsible for the correct accomplishment of the entire case-based retrieval process. Its first action is to send the query to the parsing agent that transforms it from the architectural GraphML into an ontological representation in *SL* content language to provide human-readability as well as suitability for communication among agents according to (Caire and Cabanillas, 2002).
Back at the coordination point, the query is analyzed and sent to a *retrieval container* that conducts the actual resolving of this particular query. The coordinator selects the proper retrieval strategy and conducts the setup of the container, where a *retrieval manager* and at least one *case-based retrieval agent* that operates with at least one *similarity measure* μ are mandatory units for successful processing of the retrieval stage in the case base(s) of the CBR domain model.

After finishing the actual retrieval process, the achieved result data is sent back from the container's manager to the coordinator, who forwards it to the gateway that transforms it into the proper processing format (JSON) and sends the result data object to the web interface component for displaying.

Parallel to the user query resolving process, new cases *(building designs)* are being imported into the case base(s) of the underlying CBR domain model. This is done by another essential component of the proposed MAS, the *maintainer*, whose main task is to extract cases from the BIM infrastructure's graph database and to import them into these case base(s).

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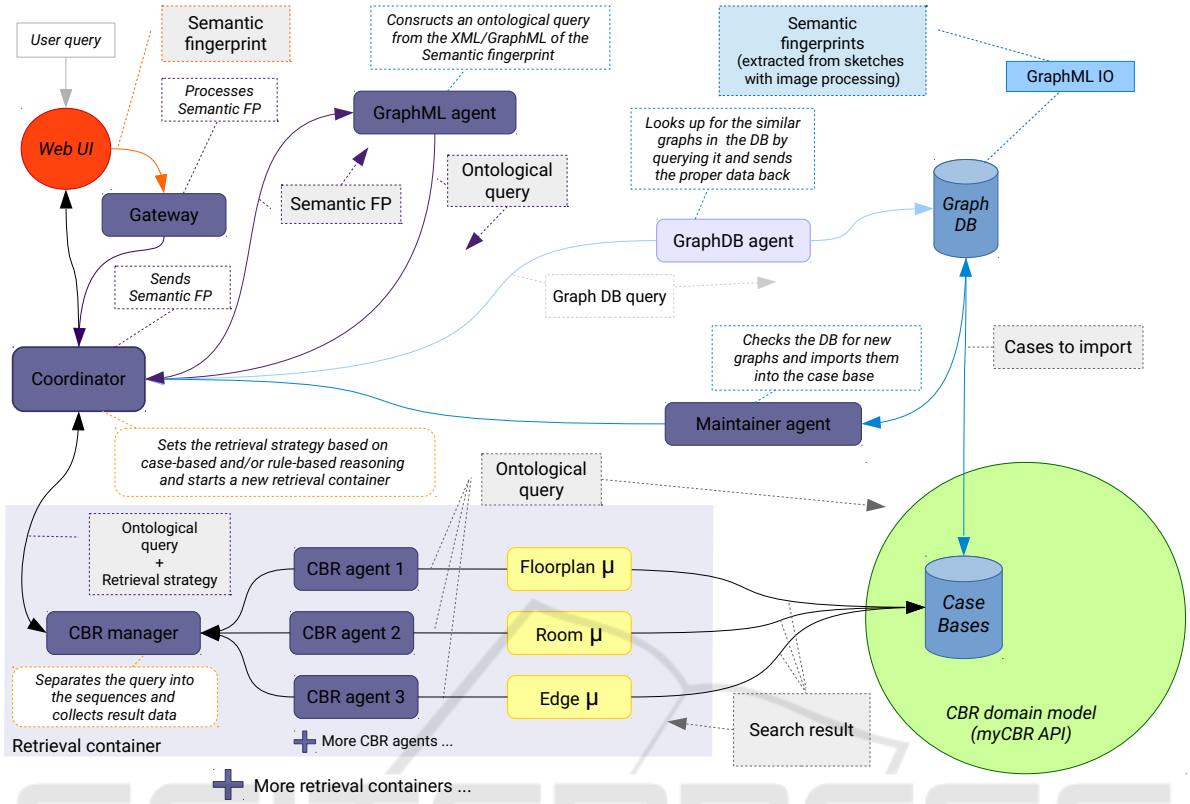


Figure 1: The system overview of MetisCBR.

4 系统概述

在本节中，我们对所提出的方案*MetisCBR*进行简要概述。整个多智能体系统的主要任务是处理由用户*(an archi-tect)*通过基于Web的查询构建界面（不属于本方案的一部分，参见第3节和第6节）发送的AGraphML格式的查询。

如图1所示，该系统采用环形架构，其中*gateway*标识用户通过Web界面提交查询后的系统入口。

在成功通过执行初始错误分析的网关代理后，查询将到达负责整个基于案例的检索过程正确执行的*coordination component*。其首要操作是将查询发送至解析代理，该代理将其从架构化的GraphML格式转换为采用*SL*内容语言的本体论表示，从而既保证人类可读性，又使其符合根据（Caire和Cabanillas，2002）所述的各代理间通信的要求。

在协调点，查询被分析并发送至一个*retrieval container*，该节点负责执行相应的操作——对该特定查询的实际求解。协调器选择适当的检索策略，并完成容器的配置；在基于案例推理（CBR）领域模型的案例库中，成功执行检索阶段所必需的组件包括一个*retrieval manager*以及至少一个与至少一个*similarity measure μ*协同工作的*case-based retrieval agent*。

在实际检索过程完成后，所得结果数据由容器管理器返回给协调器，协调器再将其转发至网关，由网关将其转换为适当的处理格式（JSON），并将结果数据对象发送至Web界面组件以供展示。

与用户查询求解过程并行，新的案例*(building designs)*正被导入到底层CBR领域模型的案例库中。这一操作由所提出的MAS的另一关键组件——*maintainer*——完成，其主要任务是从BIM基础设施的图数据库中提取案例，并将其导入这些案例库。

5 AGENT TYPES AND TASKS

This section presents detailed descriptions of the agent types that are directly or indirectly involved in the process of retrieval of similar building designs. Their tasks will be also described in detail.

5.1 Agent Categories

The categorization of agents into classified groups helps to achieve a better understanding of the entire system functionality. For our case-based multi-agent approach with retrieval as main task, we defined the following categories of agents: *case-based*, *managing*, *connecting* and *service* agents. An agent may belong to one or many categories.

Case-based agents are *directly* involved in the retrieval process of similar architectural designs *and* possess an internal reasoning and learning component with rule-based and/or case-based reasoning abilities. Retrieval executing agents of this category (see Section 5.3.2) are able to apply different similarity measures when searching for designs (or parts of it) in the case base(s) of the CBR domain model described in (Ayzenshtadt et al., 2015) (see also Sections 3 and 6).

Managing agents ensure that the entire system (including the retrieval process and building designs import into the case base(s)) is working properly, without failures, such as abruptions or sudden bottlenecks. The agents of this class monitor ongoing internal processes, share their statuses/intentions, and forward queries and results to the corresponding agents.

Connecting agents play the role of connection points between the MAS and linked external services. They make sure that the connection and communication of *MetisCBR* with these environments (e.g., the query builder interface or the graph database) is established, and query or case streams can be forwarded to the destinated system components.

Service agents are helper agents that execute tasks that do not belong directly to retrieval, management, or connection task domains. The agents of this class are involved in the accomplishment of intermediate steps of the retrieval process (e.g., query parsing).

5.2 Coordination

For the efficient coordination of the retrieval process within our system, a corresponding agent (the *coordinator*) was implemented. Its role is to manage the entire case-based retrieval stage, after the user query has been received from the gateway. Therefore, this agent belongs to both categories of *case-based* and *managing* agents, and has a number of particular tasks.

Retrieval Strategy Selection. The selection of the proper retrieval strategy is the coordinator's most essential task. It is based on the combination of the properties of the query, previous experiences, rules, and user-specified data from the query.

For example, if the user's current objective is to conduct a retrieval without much detailed settings for the search, the coordinator can derive the objective from the user-specified data (in form of a specific attribute) and select the associated retrieval strategy to resolve the query. Otherwise, the common strategy can be applied. The objective recognition and setting is an example of applying the rule-based reasoning, the first one of the reasoning abilities of the coordinator. The second one is the case-based classification of the query, where selection of the strategy is mainly the task of the learning component of the coordinator.

Learning is a significant property of an agent that aims to improve its ability to make decisions based on previous experiences. Therefore, the coordination component of our retrieval system, being such an agent, uses its case-based learning ability as an essential feature to select the proper procedure for query resolving. Each query that the coordinator gets from a user, together with the achieved results, is registered as a case with the IB2 algorithm (Aha et al., 1991) that controls the filling of the coordinator's internal case base and provides the coordinator with the learning ability. These cases are the base for the reasoning process that gets activated if the rule-based classification misses the explicit specification from the user side. During the CBR-based classification, the corresponding floor plan information of the query (e.g., room count and room types) is used to find the most similar case in the coordinator's internal case base, and apply a strategy that was used in this case.

Retrieval Container Activation. After the strategy for the retrieval has been selected, the coordinator activates a new retrieval container (see Section 5.3). The setting of the container is defined by the coordinator – it selects the retrieval agents that will execute the actual retrieval steps and starts the activity of the container. In *MetisCBR* the retrieval strategy is essential for selection of the agents – *retrieval agent types are bound to the strategy*, but *not relevant* for the selection of the strategy. The coordinator is also able to terminate one or more currently running retrieval containers. This happens when a container has finished its retrieval activities, *and* the user has finished her retrieval session. In case of unexpected interruption of the connection between the user interface and the MAS, the retrieval is continued and the results are persisted in an intermediate object.

5 种代理类型与任务

本节详细阐述了在相似建筑方案检索过程中直接或间接参与的各类智能体类型，并对其具体任务进行了详尽说明。

5.1 代理类别

将智能体划分为不同的类别有助于更深入地理解整个系统的功能。对于以检索为主要任务的基于案例的多智能体方法，我们定义了以下几类智能体：*case-based*、*manag-*、*ing*、*connecting* 和 *service* 类智能体。一个智能体可以属于某一类或多类。

Case-based agents directly 参与相似建筑方案的检索过程；*and* 具备基于规则和/或基于案例推理的内部推理与学习机制。该类检索执行代理（参见第5.3.2节）在搜索CBR领域模型（见Ayzenshtadt等，2015）的案例库时，能够应用不同的相似性度量方法（参见第3节和第6节）。

Managing agents 确保整个系统（包括检索过程以及将设计成果导入案例库）运行正常，不存在诸如中断或突发瓶颈等问题。该类智能体负责监控当前的内部流程，共享其状态与意图，并将查询及结果转发至相应的智能体。

Connecting agents 充当MAS与所链接的外部服务之间的连接点。它们确保MAS与这些环境（例如查询构建器界面或图数据库）之间的连接与通信得以建立，并使查询流或案例流能够被转发至目标系统组件。

Service agents 它们是辅助代理，负责执行不属于检索、管理或连接任务范畴的任务。此类代理参与检索过程中的中间步骤（例如查询解析）。

5.2 协调

为实现本系统内检索过程的高效协调，我们部署了一个相应的代理（*coordi- nator*）。该代理负责在网关接收用户查询后，管理整个基于案例的检索阶段。因此，该代理同时属于*case-based*类和*manag- ing*类代理，并承担多项特定任务。

检索策略的选择。选择恰当的检索策略是协调员最为重要的任务。这一选择基于查询特征、以往经验、相关规则以及用户在查询中指定的数据的综合考量。

例如，若用户当前的目标是在无需进行过多细节设置的情况下执行检索，协调器可从用户指定的数据（以特定属性的形式）中推导出该目标，并选择与之关联的检索策略来处理查询。否则，则可采用通用策略。目标识别与设定是应用基于规则推理的一个实例，这也是协调器所具备的第一种推理能力。第二种推理能力则是基于案例的查询分类，其中策略的选择主要由协调器的学习组件负责完成。

学习是智能体的一项重要特性，旨在基于以往经验提升其决策能力。因此，作为此类智能体的检索系统协调组件，将基于案例的学习能力作为核心功能，用于选择恰当的查询求解流程。协调组件从用户处接收到的每一条查询及其所得结果，均以案例形式记录，并由IB2算法（Aha等，1991）负责管理协调组件内部案例库的构建，从而赋予其学习能力。这些案例构成了推理过程的基础；当基于规则的分类未能获得用户明确指定时，该推理过程即被激活。在基于案例的推理过程中，查询所对应的平面布局信息（如房间数量及房间类型）被用于在协调组件的内部案例库中查找最相似的案例，并应用该案例中所采用的策略。

检索容器激活。在选定检索策略后，协调器会激活一个新的检索容器（参见第5.3节）。容器的配置由协调器确定——它选择将执行实际检索步骤的检索代理，并启动该容器的活动。在*MetisCBR*中，检索策略对于代理的选择至关重要——*retrieval agent types are bound to the strategy*，但*not relevant*则用于策略的选择。协调器还能够终止一个或多个当前正在运行的检索容器。这种情况发生在：当某个容器已完成其检索任务时，或者*and*用户已结束其检索会话时。若用户界面与多智能体系统之间的连接发生意外中断，检索将继续进行，且结果将被持久化存储于一个中间对象中。

Forwarding of the Query Modifications. If it is the user's wish to modify the current query during the ongoing retrieval, the coordinator receives the modified query from the gateway agent and forwards the changes while notifying the manager of the corresponding container about these changes.

Sending of the Results to the Gateway. When the actual retrieval phase is finished, the coordinator receives the results from the manager of the container and provides the gateway agent with these results.

Assignment of the Query Parsing Task. The coordinator assigns the *GraphML agent* (see Section 5.6) the task of parsing the current query, which it gets back transformed as an ontological query in SL content language format.

Receiving the Import Status. From the maintainer agent the coordinator receives the current status of the extraction/import process and coordinates the currently running retrieval tasks with respect to this status (see also Section 5.5).

5.2.1 Coordination Techniques

The efficient accomplishment of the above described tasks requires possession of corresponding coordination techniques. The coordination component of our retrieval system is developed with the integration of combination of some of the coordination scenarios described in (Nwana et al., 1996).

Organizational Structuring is applied when the coordinator creates and destroys retrieval containers and assigns strategies to them. It is the main coordination technique in *MetisCBR*.

Contracting (more precisely, a sub-feature of contracting, the *sub-distribution* of the tasks) is applied when a manager of a retrieval container assigns parts of the query to be resolved by corresponding agents.

Centralised Multi-agent Planning is activated when the maintainer sends an import status to the coordinator (see Section 5.5)

5.3 Retrieval Container

In a retrieval system, where concurrent queries are expected to be a usual case, a big advantage is to implement a structure that is able to handle each of the queries separately to achieve more efficient processing. For the actual building design retrieval step of our approach, we use a *container-based* structure, where each *retrieval container* during its activity can be seen as a completely autonomous multi-agent sub-system that is nested inside a parent MAS, and can only be terminated if its retrieval process and the user session are both finished.

The working cycle of a retrieval container starts with receiving of an ontological query from the coordinator and ends after the retrieval of the assigned query is finished. Between those two steps, a case-based resolving of the query takes place inside the container. Thus, every container has three goals to complete during its working cycle: *receiving* of the query, *resolving* of the query by means of applying CBR methods, and *replying* to the coordinator with a message that contains the achieved results.

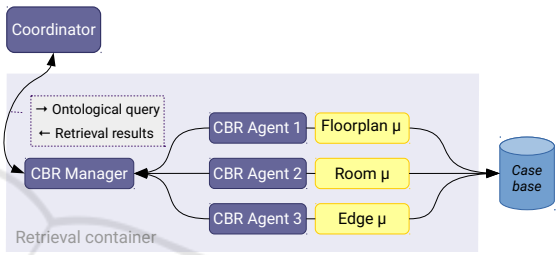


Figure 2: Exemplary configuration of a retrieval container.

An exemplary configuration of a single retrieval container can be seen in Figure 2. Every container consists of a mandatory *CBR manager* – an agent that manages resolving of the current query – and at least one mandatory *CBR retrieval agent* – an entity that is responsible for the actual search for similar cases. The following sections provide a detailed description of both agent types.

5.3.1 CBR Manager

As mentioned above, the CBR manager is a mandatory agent inside an active retrieval container. It organizes the current case-based retrieval of an ontological query that was received from the coordinator. Therefore, this agent belongs to the category of *managing agents* (see Section 5.1), and was primarily created to take load off the coordinator, that can in turn concentrate on its strategic and system managing tasks.

The tasks of the manager include two primary classes: *communication with the coordinator* and *query separation*. During the communication with the coordinator, (i.e., receiving an ontological query and sending the results back), the manager completes *receiving* and *replying* steps of the container goals. The separation process cuts the given query into the number of parts according to the selected retrieval strategy and forwards these parts to the corresponding agents.

查询修改的转发。若用户希望在检索过程中对当前查询进行修改，协调器将从网关代理处接收修改后的查询，并在将这些修改转发的同时，通知相应容器的管理器有关这些修改。

将结果发送至网关。当实际检索阶段结束后，协调器从容器管理器处接收结果，并将其提供给网关代理。

查询解析任务的分配。协调器将当前查询的解析任务分配给*GraphML agent* (参见第5.6节)，该查询随后被转换为以SL内容语言格式表示的本体查询并返回。

接收导入状态。协调器从维护代理处获取当前的抽取/导入进程状态，并根据该状态对正在运行的检索任务进行协调（参见第5.5节）。

5.2.1 协调技术

上述任务的高效完成需要具备相应的协调技术。我们检索系统的协调组件是在综合（Nwana等，1996）中所描述的部分协调场景的基础上开发的。

当协调器创建并销毁检索容器并向其分配策略时，会应用组织结构化机制。这是*MetisCBR*中的主要协调技术。

当检索容器的管理者将查询任务的部分内容分配给相应的代理进行处理时，便会应用委托机制（更准确地说，是委托这一子功能，即任务的*sub-distribution*）。

当维护人员向协调器发送导入状态时（参见第5.5节），集中式多智能体规划机制即被激活。

5.3 检索容器

在检索系统中，由于并发查询是常见情形，采用能够独立处理每一条查询的结构具有显著优势，从而实现更高效的处理。对于本方法中的实际建筑方案检索步骤，我们采用了一种*container-based*结构，其中在其运行期间，每个*retrieval container*均可被视为一个完全自治的多智能体子系统，该子系统嵌套于其父级MAS之中，并且仅当其检索过程与用户会话均已完成时方可终止。

检索容器的工作周期始于接收来自协调器的本体查询，终于所分配查询的检索任务完成。在这两个步骤之间，容器内部会进行基于案例的查询求解。因此，每个容器在其工作周期内需完成三项任务：*receiving* 查询、*resolving* 通过应用基于案例的推理方法对查询进行求解，以及 *replying* 向协调器发送包含求解结果的消息。

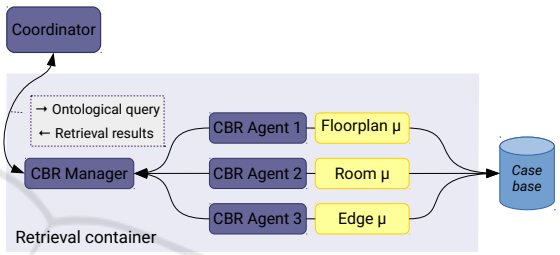


图2：检索容器的示例配置。

单个检索容器的典型配置如图2所示。每个容器均由一个必选的*CBR manager*——负责管理当前查询的解析——以及至少一个必选的*CBR retrieval agent*——负责执行相似案例的实际检索——组成。以下各节将对这两种代理类型进行详细说明。

5.3.1 CBR 管理器

如上所述，CBR管理器是活动检索容器中的一个强制性代理。它负责组织对来自协调器的本体查询的当前基于案例的检索。因此，该代理属于*managing agents* (参见第5.1节)这一类别，其主要设计目的是减轻协调器的负担，使其能够专注于自身的战略任务和系统管理工作。

管理器的任务包括两大类：*communication with the coordinator* 和 *query separation*。在与协调器进行通信的过程中（即接收本体查询并返回查询结果），管理器会完成容器目标中的 *re-*、*ceiving* 和 *replying* 步骤。分离过程根据所选的检索策略将给定的查询划分为若干个子部分，并将这些子部分转发至相应的代理。

The secondary class of tasks of the manager consists of helper operations, which are accomplished only if required by the strategy as well. Those helper operations can include some computational tasks, such as determination of rooms and room connections that originate from the same floor plan.

5.3.2 CBR Retrieval Agents

CBR retrieval agents execute the last and most important part of the retrieval process – they apply CBR retrieval methods for determination of the most similar cases to the query or its parts in the case base(s) of the underlying *attribute-value based* CBR domain model (described in detail in (Ayzenshtadt et al., 2015)). To achieve this goal, they use similarity functions prepared for each agent, special *query types*, and an internal reasoning mechanism to deal with incoming queries from the manager. CBR retrieval agents belong to the *case-based agents*, where several agent types of this class currently exist.

Floor plan Agent. This agent's aim is to search for cases that contain *floor plan meta data* from the AGraphML specification and additional related information such as room count and room types. Floor plan agent works with following query types:

- `Floorplan Query` – every attribute of the meta data and additional data will be taken into account for the similarity measurement.
- `Floorplan ID List Query` – returns cases, if their corresponding IDs exist in a defined ID list.

Room Agent. To find similar rooms to a given one this agent uses every room attribute from the domain model. Its only query type is currently `Room Query` that accomplishes this similarity computation task.

Edge Agent. This retrieval agent takes every edge attribute into account to compute a similarity measure between two room connections. The corresponding query type is the `Edge Query`.

5.3.3 Learning of CBR Retrieval Agents

CBR retrieval agents are able to learn from previous queries in the same manner as the coordinator does. Each of the retrieval agents possesses an internal reasoning mechanism, where a special *internal* CBR domain model exists for each retrieval agent's knowledge base. Special attributes are determined to hold information about previous queries the agent has resolved (e.g., room count from the query and from the found case(s) for the *Room agent*).

5.3.4 Container Identification

Retrieval container can be identified via an UUID as defined in the RFC 4122. The coordinator assigns an UUID to each container it creates, the same ID is also temporarily assigned to the query that the container is working with to be able to forward modifications to the associated container.

5.4 Gateway

As described above in the overview section, the *gateway agent* is an entry point to the MAS from the user side, and an interface for forwarding queries and results in corresponding directions. The gateway agent belongs to the category of *connecting agents* (see Section 5.1).

Gateway agents of such type are common in systems with structure similar to ours. For example, in (Greenwood et al., 2005) a gateway agent is a part of a component that is used for the OWL-S-based semantic interaction between a web client from an external web service environment and an agent environment.

The activity area of our gateway agent is divided into two main sections: *accepting* of user queries and forwarding them to the coordinator, and *receiving* results from the coordinator and forwarding them to the web user interface. Both task sections are controlled by special agent behaviors (see Figure 3).

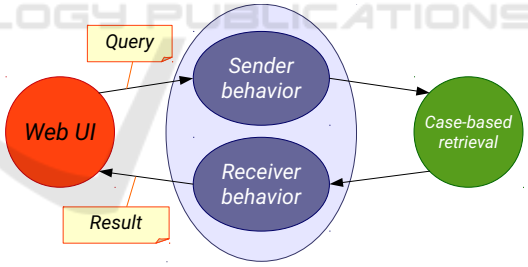


Figure 3: The behaviors structure of the gateway agent.

Sending a formalized and validated user query in AGraphML format to the coordinator is the main purpose of creating of the *Sender behavior*. During the validation of the query the gateway agent traverses through the data in order to find formal errors, such as missing meta data or false (or missing) room or room connection definitions. Empty queries will not be forwarded to the coordinator. Incomplete queries, where data is fragmentary or only partly missing, will be forwarded to the coordinator, however, missing attributes will not be used for the retrieval.
Receiving result data from the coordinator and delivering it to the web-based user interface is the activity

管理器的第二类任务由辅助操作组成，这些辅助操作仅在策略需要时才会执行。此类辅助操作可能包括一些计算任务，例如根据同一楼层平面图确定房间及其连通关系。

5.3.2 CBR检索代理

CBR retrieval agents 执行检索过程的最后且最为重要的环节——它们运用案例推理（CBR）检索方法，以确定与查询或其组成部分在底层 *attribute-value based* CBR域模型的案例库中最为相似的案例（详见Ayzenshtadt等，2015）。为实现这一目标，它们使用为每个代理预先设定的相似度函数、特定的*query types*，以及一种用于处理来自管理器的传入查询的内部推理机制。CBR检索代理隶属于*case-based agents*，该类目前存在多种代理类型。

平面图代理。该代理的目标是从AGraphML规范中检索包含*floor plan meta data*的案例，并获取相关的附加信息，如房间数量和房间类型。平面图代理支持以下查询类型:

- 楼层平面查询——在相似度计算中，将综合考虑元数据及附加数据的每一项属性。
- 楼层平面图ID列表查询——若指定的ID列表中存在相应ID，则返回对应的案例。

房间代理。为查找与给定房间相似的房间，该代理会使用领域模型中的所有房间属性。目前，其唯一的查询类型为房间查询，用于完成这一相似度计算任务。

边缘代理。该检索代理会综合考虑每条边的属性，以计算两个房间连接之间的相似度量。相应的查询类型为边缘查询。

5.3.3 基于案例的推理检索代理的学习

基于案例的推理检索代理能够以与协调器相同的方式从先前的查询中学习。每个检索代理都配备了一个内部推理机制，其中为每个检索代理的知识库构建了一个专门的CBR领域模型。此外，还定义了若干特殊属性，用于存储该代理已解决的先前查询的相关信息（例如，来自查询及所检索到的案例中的房间数量）。

5.3.4 容器识别

检索容器可通过RFC 4122中定义的UUID进行标识。协调器为其创建的每个容器分配一个UUID，并将同一ID临时分配给该容器正在处理的查询，以便能够将修改操作转发至相应的容器。

5.4 网关

如概述部分所述，*gate- way agent* 是用户侧进入MAS 的入口，同时也是在相应方向上转发查询与结果的接口。该网关代理属于 *connecting agents* (类别，详见第 5.1 节)。

此类网关代理在与我们系统结构类似的系统中较为常见。例如，在（Greenwood等，2005）中，网关代理是某个组件的一部分，用于实现来自外部Web服务环境的Web客户端与代理环境之间的基于OWL-S的语义交互。

我们的网关代理的活动区域分为两个主要部分：*accepting*接收用户查询并将其转发至协调器，以及*receiving*接收来自协调器的结果并将其转发至Web用户界面。这两个任务模块均由特定的代理行为进行控制（见图3）。

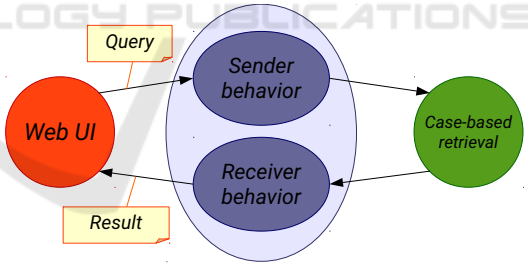


图3：网关代理的行为结构。

以AGraphML格式向协调器发送经过形式化且已验证的用户查询，是创建*Sender behavior*的主要目的。在查询验证过程中，网关代理会遍历数据，以查找诸如元数据缺失、房间或房间连接定义错误（或缺失）等形式化错误。空查询将不会被转发至协调器。对于数据片段化或仅部分缺失的不完整查询，虽会被转发至协调器，但缺失的属性将不会用于检索过程。

从协调器接收结果数据并将其传递至基于Web的用户界面，这一过程即为该活动。

field of the *Receiver behavior* of the gateway agent. Similar to the process of the sender behavior, the result data (in SL content language) will be validated as well, following by the transformation into the JSON format (plain HTML is also possible). It is optional if results will be returned in sequences or at once.

5.5 Maintainer

Maintainer is an agent that is responsible for the correct functionality of the extraction and import of the graph-based designs from the BIM infrastructure's graph database into the case base(s) of the underlying case-based domain model of the system. The maintainer belongs to the class of *managing agents*.

During the three-step case base(s) update process the maintainer executes tasks defined for each step. In the first phase it queries the graph database for new graph representations of building designs. In the second phase, the maintainer requests the RESTful service (implemented in the BIM infrastructure) for each graph of the previous result set and gets back an AGraphML-formatted graph. In the last step, the building graph is imported to the case base(s) of the domain model with a special import function.

For the essential workflow of the system, the maintainer is also able to set and share an *import status*, which currently can be one of the following: `Import Ready`, `Import Active`, `Import Paused`, and `Import Done`.

The coordinator receives the status and may execute actions according to it. For example, if the system performance is critical the coordinator may stop some retrieval containers or request to stop recently started extraction/import process to allow for a better system resource sharing.

5.6 GraphML Agent

As previously mentioned (see Section 5.2), the *GraphML agent* is the system entity that is responsible for parsing of the incoming architectural GraphML queries into ontological queries in *SL* content language format. The ontological queries get their name from the underlying communication ontology that defines the communication vocabulary among agents. By means of applying an ontology to the communication, the ontological query becomes part of the message content as demonstrated in (Caire and Cabanillas, 2002).

6 SYSTEM EVALUATION

For the above mentioned basic research project *Metis*, a working prototype of the proposed retrieval system *MetisCBR* was developed to demonstrate its suitability for the "real-world" application. Java language based frameworks *JADE* (for the implementation of the described agent system) and *myCBR* were used to develop the prototype. It is also implemented as one of the sub-systems of the BIM infrastructure.

The prototype works with the BIM infrastructure's content management system (that contains *graphical* representations of building designs) and from it derived graph database that contains every building design as *graph representation* that is then extracted and imported as specified in Section 5.5. In Section 3 mentioned web-based user interface is planned to be used as the main *graphical* architectural GraphML query builder for the retrieval system. Furthermore, the proposed system and the software prototype are part of the master thesis (Ayzenshtadt, 2015).

For the validation and demonstration of the abilities of the approach, an evaluation of the system was conducted, that is completely described in (Ayzenshtadt et al., 2015). The evaluation process included the querying of the system with an exemplary query consisting of 5 rooms and 6 connecting edges with two different similarity measures: with and without attribute weighting.

The *MAS task area* of the evaluation should show that *MetisCBR* is already able to handle search requests and return results according to the similarity measures.

Considering the early phase of the system implementation, the agents were able to properly handle the user queries. The coordinator was able to recognize a new query, to create a corresponding container, and to assign the strategy and the query type. The manager of the container was able to recognize if it is required to separate the query, and then assigned the corresponding parts to the CBR retrieval agent(s). The retrieval agent(s) retrieved the case base with the similarity measure that was set for the current query (manually, on this stage of development). The manager collected the results and sent them to the coordinator that then forwarded them to the gateway agent. A couple of minor technical problems occurred during the import process, but could be traced by inspecting the consistency service of the BIM infrastructure.

The results of the evaluation of the CBR task area, including the information about retrieval strategies, similarity measures, building design dataset used, average and highest similarity, and the similarity distribution, are described in (Ayzenshtadt et al., 2015).

网关代理的*Receiver behavior*字段。与发送方行为的处理过程类似，结果数据（采用SL内容语言）也将经过验证，随后被转换为JSON格式（亦可采用纯HTML格式）。结果数据可以选择按顺序返回，也可一次性全部返回。

5.5 维护者

Maintainer 该代理负责确保从BIM基础设施的图数据库中将基于图的设计正确地提取并导入至系统底层基于案例的领域模型的案例库中。该维护者属于*managing agents*类。

在三步案例库更新过程中，维护人员依次执行各步骤中定义的任务。在第一阶段，维护人员向图数据库查询建筑方案的新图表示；在第二阶段，针对前一阶段结果集中的每一张图，维护人员通过BIM基础设施中实现的RESTful服务请求相应的AGraphML格式图；在最后一步，利用专用的导入功能将建筑图导入到领域模型的案例库中。

对于系统的核心工作流，维护人员还可以设置并共享一个状态标识*import status*，当前该标识可取以下值之一：导入就绪、导入进行中、导入已暂停以及导入已完成。

协调器接收系统状态，并可根据该状态执行相应操作。例如，当系统性能处于临界状态时，协调器可能会停止部分检索容器，或请求终止近期启动的抽取/导入进程，以实现更优的系统资源分配。

5.6 GraphML 代理

如前所述（参见第5.2节），*GraphML agent*是负责将传入的架构化GraphML查询解析为以*SL*内容语言格式表示的本体论查询的系统实体。这些本体论查询得名于其底层的通信本体，该本体定义了各智能体之间的通信词汇。通过将本体应用于通信过程，本体论查询便成为消息内容的一部分，如(Caire和Cabanillas, 2002)所示。

6 系统评估

针对上述基础研究项目*Metis*，开发了所提出的检索系统*MetisCBR*的原型，以验证其在“真实场景”中的适用性。在原型开发过程中，采用了基于Java语言的框架*JADE* (来实现所述的代理系统)和*myCBR*。该原型亦作为BIM基础设施的子系统之一加以实现。

该原型与BIM基础设施的内容管理系统（其中包含建筑模型的*graphical*表示）以及由此派生的图数据库协同工作，该图数据库以*graph representation*形式存储每一项建筑模型，并按照第5.5节的规定进行提取与导入。在第3节中，计划将所提及的基于Web的用户界面用作检索系统的主要*graphical*架构化GraphML查询构建工具。此外，所提出的系统及软件原型亦为硕士论文（Ayzenshtadt，2015）的一部分。

为了验证并展示该方法的有效性，对系统进行了评估，其详细内容见（Ayzenshtadt 等，2015）。评估过程包括使用一个示例查询对系统进行测试，该查询由5个房间和6条连接边组成，并采用了两种不同的相似度度量：一种考虑属性权重，另一种则不考虑。

评估中的*MAS task area*应表明，*MetisCBR*已能够处理搜索请求，并根据相似性度量返回结果。

鉴于系统实施的早期阶段，各代理能够妥善处理用户查询。协调器能够识别新查询、创建相应的容器，并为其分配策略与查询类型。容器管理器能够判断是否需要对查询进行拆分，并将相应部分分配给一个或多个基于案例推理的检索代理。检索代理根据为当前查询设定的相似性度量（在本开发阶段由人工配置）从案例库中进行检索。管理器汇总检索结果并将其发送至协调器，再由协调器转发至网关代理。在导入过程中出现了一些 minor 的技术问题，但通过检查 BIM 基础设施的一致性服务得以定位。

关于案例推理任务领域的评估结果，包括检索策略、相似性度量、所使用的建筑设计数据集、平均相似度与最高相似度，以及相似度分布等信息，详见（Ayzenshtadt 等，2015）。

7 CONCLUSION

In this paper we presented *MetisCBR*, a distributed case-based approach for the retrieval of similar building designs in the corresponding case base(s). The system consists of different agent types, where the coordinator controls the retrieval process and applies rule-based or case-based reasoning to select the proper retrieval strategy, retrieval containers execute concurrently the actual case-based retrieval processes, the gateway agent connects the system with an external service (web-based query builder) and the maintainer agent manages the import of new cases into the case base(s) of the underlying CBR domain model. Service agents contribute to the smooth run of the system by executing small intermediate tasks.

The future work includes the full integration of the graphical user interface to the presented MAS. Index-based retrieval method that queries the graph database directly is also planned to be added, a new *GraphDB* agent will be integrated to the MAS to accomplish this alternative retrieval task. This will also allow for evaluating and hopefully deeply integrating these different kinds of retrieval approaches.

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7 结论

在本文中，我们提出了*MetisCBR*，这是一种用于在相应案例库中检索相似建筑设计的分布式基于案例的方法。该系统由多种类型的代理组成：协调器负责控制检索过程，并采用基于规则或基于案例的推理来选择合适的检索策略；检索容器并行执行实际的基于案例的检索过程；网关代理将系统与外部服务（基于Web的查询构建器）相连接；维护代理则管理新案例向底层CBR领域模型案例库的导入。服务代理通过执行一些小型的中间任务，确保系统的平稳运行。

未来的工作包括将所提出的多智能体系统（MAS）与图形用户界面进行全面集成。此外，还计划引入一种直接查询图数据库的基于索引的检索方法，并将一个新的*GraphDB*智能体集成到该系统中，以实现这一替代性检索任务。这也将有助于对这些不同类型的检索方法进行评估，并有望实现其深度整合。

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